

Tokenization

```
In [1]: import nltk
nltk.download('punkt')
nltk.download('wordnet')
nltk.download('averaged_perceptron_tagger')
nltk.download('stopwords')
from nltk import sent_tokenize
from nltk import word_tokenize
from nltk.corpus import stopwords
```

```
[nltk_data] Downloading package punkt to
[nltk_data]   C:\Users\RBS\AppData\Roaming\nltk_data...
[nltk_data]   Unzipping tokenizers\punkt.zip.
[nltk_data] Downloading package wordnet to
[nltk_data]   C:\Users\RBS\AppData\Roaming\nltk_data...
[nltk_data]   Unzipping corpora\wordnet.zip.
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data]   C:\Users\RBS\AppData\Roaming\nltk_data...
[nltk_data]   Unzipping taggers\averaged_perceptron_tagger.zip.
[nltk_data] Downloading package stopwords to
[nltk_data]   C:\Users\RBS\AppData\Roaming\nltk_data...
[nltk_data]   Unzipping corpora\stopwords.zip.
```

```
In [2]: text='Real madrid is set to win the UCL for the season . Benzema might win Balon do
```

```
In [3]: tokens_sents = nltk.sent_tokenize(text)
print(tokens_sents)
```

```
['Real madrid is set to win the UCL for the season .', 'Benzema might win Balon dor', 'Salah might be the runner up']
```

```
In [4]: tokens_words = nltk.word_tokenize(text)
print(tokens_words)
```

```
['Real', 'madrid', 'is', 'set', 'to', 'win', 'the', 'UCL', 'for', 'the', 'season', '.', 'Benzema', 'might', 'win', 'Balon', 'dor', '.', 'Salah', 'might', 'be', 'the', 'runner', 'up']
```

```
In [5]: from nltk.stem import PorterStemmer
from nltk.stem.snowball import SnowballStemmer
from nltk.stem import LancasterStemmer
```

```
In [9]: stem=[]
for i in tokens_words:
    ps = PorterStemmer()
    stem_word= ps.stem(i)
    stem.append(stem_word)
print(stem)
```

```
['real', 'madrid', 'is', 'set', 'to', 'win', 'the', 'ucl', 'for', 'the', 'season', '.', 'benzema', 'might', 'win', 'balon', 'dor', '.', 'salah', 'might', 'be', 'the', 'runner', 'up']
```

Lemmatization

```
In [10]: import nltk
         from nltk.stem import WordNetLemmatizer
         lemmatizer = WordNetLemmatizer()
```

```
In [11]: lemmatized_output = ' '.join([lemmatizer.lemmatize(w) for w in stem])
         print(lemmatized_output)
```

real madrid is set to win the ucl for the season . benzema might win balon dor . salah might be the runner up

```
In [12]: leme=[]
         for i in stem:
             lemetized_word=lemmatizer.lemmatize(i)
             leme.append(lemetized_word)
         print(leme)
```

['real', 'madrid', 'is', 'set', 'to', 'win', 'the', 'ucl', 'for', 'the', 'season',
'.', 'benzema', 'might', 'win', 'balon', 'dor', '.', 'salah', 'might', 'be', 'the',
'runner', 'up']

Part of Speech Tagging

```
In [13]: print("Parts of Speech: ",nltk.pos_tag(leme))
```

Parts of Speech: [('real', 'JJ'), ('madrid', 'NN'), ('is', 'VBZ'), ('set', 'VBN'),
(('to', 'TO'), ('win', 'VB'), ('the', 'DT'), ('ucl', 'NN'), ('for', 'IN'), ('the', 'D
T'), ('season', 'NN'), ('.', '.'), ('benzema', 'NN'), ('might', 'MD'), ('win', 'V
B'), ('balon', 'NN'), ('dor', 'NN'), ('.', '.'), ('salah', 'NN'), ('might', 'MD'),
(('be', 'VB'), ('the', 'DT'), ('runner', 'NN'), ('up', 'RP')]

Stop Word

```
In [14]: sw_nltk = stopwords.words('english')
         print(sw_nltk)
```

```
[ 'i', 'me', 'my', 'myself', 'we', 'our', 'ours', 'ourselves', 'you', "you're", "yo
u've", "you'll", "you'd", 'your', 'yours', 'yourself', 'yourselves', 'he', 'him', 'h
is', 'himself', 'she', "she's", 'her', 'hers', 'herself', 'it', "it's", 'its', 'itse
lf', 'they', 'them', 'their', 'theirs', 'themselves', 'what', 'which', 'who', 'who
m', 'this', 'that', "that'll", 'these', 'those', 'am', 'is', 'are', 'was', 'were',
'be', 'been', 'being', 'have', 'has', 'had', 'having', 'do', 'does', 'did', 'doing',
'a', 'an', 'the', 'and', 'but', 'if', 'or', 'because', 'as', 'until', 'while', 'of',
'at', 'by', 'for', 'with', 'about', 'against', 'between', 'into', 'through', 'durin
g', 'before', 'after', 'above', 'below', 'to', 'from', 'up', 'down', 'in', 'out', 'o
n', 'off', 'over', 'under', 'again', 'further', 'then', 'once', 'here', 'there', 'wh
en', 'where', 'why', 'how', 'all', 'any', 'both', 'each', 'few', 'more', 'most', 'ot
her', 'some', 'such', 'no', 'nor', 'not', 'only', 'own', 'same', 'so', 'than', 'to
o', 'very', 's', 't', 'can', 'will', 'just', 'don', "don't", 'should', "should've",
'now', 'd', 'll', 'm', 'o', 're', 've', 'y', 'ain', 'aren', "aren't", 'couldn', "cou
ldn't", 'didn', "didn't", 'doesn', "doesn't", 'hadn', "hadn't", 'hasn', "hasn't", 'h
aven', "haven't", 'isn', "isn't", 'ma', 'mightn', "mightn't", 'mustn', "mustn't", 'n
eedn', "needn't", 'shan', "shan't", 'shouldn', "shouldn't", 'wasn', "wasn't", 'were
n', "weren't", 'won', "won't", 'wouldn', "wouldn't"]
```

```
In [15]: words = [word for word in text.split() if word.lower() not in sw_nltk]
          new_text = " ".join(words)
          print(new_text)
```

Real madrid set win UCL season . Benzema might win Balon dor . Salah might runner